

📁 1. ****Robust Presentation Attack Detection (PAD) against Virtual Camera Injection on Android****

****Objective**:**

Design and evaluate a PAD system that resists spoofing attacks using virtual camera apps (e.g., DroidCam, OBS, etc.) to inject pre-recorded or synthetic biometric streams.

****Improvements added**:**

- * Systematically collect a dataset of ****real vs. injected face streams**** using both physical and virtual cameras.
- * Benchmark existing open-source liveness detection models on this dataset.
- * Propose ****novel temporal or optical flow–based cues**** to distinguish real from injected streams.
- * Implement a prototype liveness checker in an Android environment using ****CameraX or OpenCV****.
- * Publish benchmark results and failure cases to emphasize scientific contribution.

****Expected Deliverables**:**

- * Android PoC app with liveness detection.
- * Labeled dataset (real/virtual).
- * Evaluation metrics (EER, FAR, FRR).
- * Technical report with failure analysis.

📁 2. ****Behavioral Biometrics for Stream Authenticity Detection under Adversarial Injection****

****Objective**:**

Explore the use of ****behavioral biometrics**** (micro-movements, blink rate, attention drift) to detect when a biometric stream has been ****injected rather than recorded live****.

****Improvements added**:**

- * Use face landmark tracking (MediaPipe, Dlib) to extract fine-grain behavioral signatures.
- * Build a binary classifier to learn ****live vs injected behavior over time****.
- * Simulate attacks with scripted head movements, gaze patterns, and blinking sequences.
- * Propose a lightweight mobile inference solution for real-time stream classification.

****Expected Deliverables**:**

- * Behavioral biometric dataset (live/injected).
- * ML model for stream authentication.
- * Visualization tools showing pattern deviations.
- * Deployment-ready SDK component or Python API.

□ 3. **Adaptive Multimodal Liveness Detection: Fusing Face, Voice & Context against Stream Injection**

****Objective**:**

Design a ****multi-factor liveness system**** (face + voice + phone context) that adapts dynamically based on environment and detects inconsistencies in spoofed streams.

****Improvements added**:**

- * Fuse visual, audio, and phone sensors (light, proximity) for robust context.
- * Use decision-level or feature-level fusion of face/voice liveness.
- * Detect artifacts in ****lip-sync mismatch****, light inconsistencies, etc.
- * Evaluate how injected videos fail to spoof all modalities simultaneously.

****Expected Deliverables**:**

- * Multimodal fusion model.
- * Demonstration app on Android with adversarial test cases.
- * Confusion matrix and ROC analysis.
- * Field evaluation (low light, noisy background, etc.).